

Introduction.

运动学 — 动力学, 位控 — 力控

Joint { Revolute
prismatic

Drive { direct drive
gear / belt transmission

Sensor { Encoder for position
tachometer for velocity

第一部分

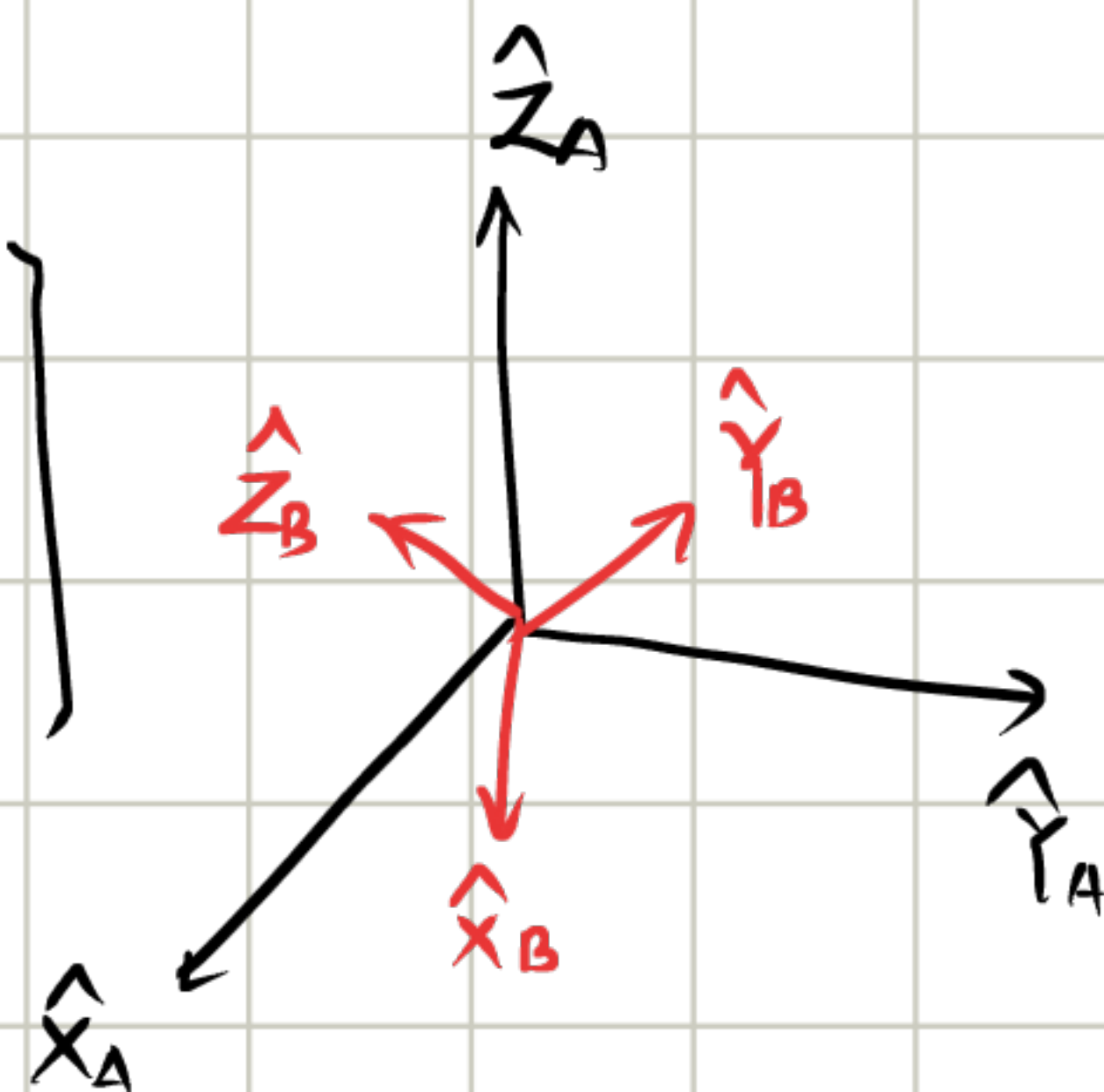
Kinematics treat motion without regard force

m_o : end effector 的自由度个数: n 关节角数量

degrees of redundancy = $n - m_o$

rotation matrix:

$${}^A_B R = [{}^A\hat{x}_B, {}^A\hat{y}_B, {}^A\hat{z}_B], \quad {}^A\hat{x}_B = \begin{bmatrix} \hat{x}_B \cdot \hat{x}_A \\ \hat{x}_B \cdot \hat{y}_A \\ \hat{x}_B \cdot \hat{z}_A \end{bmatrix}$$
$${}^A_B R = {}^B_A R^T$$



Mapping between frames

translate: ${}^A P = {}^B P + {}^A P_{B_ori}$

rotate: ${}^A P = {}^A R_B {}^B P$

general frame $\begin{pmatrix} {}^A R_B & {}^A P_{B_ori} \\ 0 & 1 \end{pmatrix}$

translate 是可交换的, 而 rotation 不可交换.

Change description from frame to frame

$${}^A P = {}^A R_B {}^B P + {}^A P_{B_ori} \Rightarrow {}^A P = {}^A T_B {}^B P \text{ (使用齐次坐标)}$$

Axis-angle: $R_k(\theta) = P_2 = R_k(\theta) P_1$

但求于 k 是任意的, 通常处理为分别对三个轴旋转

$${}^A P = {}^A T_B {}^B T_C P$$

${}^A T_B$ 为在 frame A 中看 frame B

$${}^A T_C = {}^A T_B {}^B T_C$$

n, o, a, p 表示法 (其实就是齐次矩阵)

approach / orientation / normal / position

后乘法与前乘法: 前乘用于转换到统一坐标系下 (世界坐标系)

后乘是对于连杆结构, 相对于父 joint 的坐标系

general rotation transformation: Axis-angle to rotation matrix

对于特殊轴 k

① find a coordinate frame C : k 是 C 的 z 轴 $k = (a_x, a_y, a_z)$

$k = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$, C 的 z 轴是 (a_x, a_y, a_z)

即对原坐标系的 k 就是对 C 的 z 于是 $R_k(\theta) = R_{Cz}(\theta)$

② $T = CX$ ($X = C^{-1}T$), $C = \begin{pmatrix} n & o & a & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$R_k(\theta)T = C(R_z(\theta)X) = CR_z(\theta)C^{-1}T \Rightarrow \boxed{R_k(\theta)} = CR_z(\theta)C^{-1}$$
$$= \begin{pmatrix} a_x^2(1-\cos\theta) + \cos\theta & a_x a_y(1-\cos\theta) - a_z \sin\theta & \dots & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \text{ 与 } n, o \text{ 无关}$$

于是对于连杆, 可以知道对每个坐标系的 哪个轴旋转任意角 的 R

Rotation matrix to axis angle: 已知 $R = \begin{pmatrix} n & o & a & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$\text{Trace}(R) = n_x + o_y + a_z + 1$$

$$= 2 + 2\cos\theta$$

$$\cos\theta = \frac{1}{2}(n_x + o_y + a_z - 1)$$

同时 $\sin\theta = \pm \frac{1}{2} \sqrt{(a_z - o_y)^2 + (a_x - n_z)^2 + (n_y - a_x)^2}$, 于是可以解出 k_x, k_y, k_z

Euler Angle Euler $(\phi, \theta, \psi) = R_z(\phi) R_y(\theta) R_x(\psi)$ (Z, Y, X)

也可以是 (z, y, x), (y 轴 0, 万象节死锁)
 roll (pitch (yaw 新的 z 轴

X-Y-Z Fixed Angles: RPY, 固定原有坐标系, 分别绕 X, Y, Z 轴

转三次, ${}^A_B R = R_z(\alpha) R_y(\beta) R_x(\gamma)$

RPY $(\phi, \theta, \psi) = R_z(\phi) R_y(\theta) R_x(\psi)$, 对 pitch 的扰动难度大 (坐船)

表示上与 Euler Angles 类似, 同时有 RPY $(\phi, \theta, \psi) = \text{Euler}(\phi, \theta, \psi)$

但这里是对世界坐标系; (为了理解, 可以将 Euler 看作右乘, RPY 看作左乘)

对于万象节死锁, 对前面的 z 与后面的 z 角度需要有特殊规定

表示法不能解决问题, 但可以让答案更清晰

Euler Parameters ($\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4$)



$$\begin{aligned}\epsilon_1 &= W_x \sin \frac{\theta}{2} \\ \epsilon_2 &= W_y \sin \frac{\theta}{2} \\ \epsilon_3 &= W_z \sin \frac{\theta}{2} \\ \epsilon_4 &= \cos \frac{\theta}{2}\end{aligned}$$

from axis-angle to euler parameter

from ${}^A_B R$ find ϵ :

Lemma: for all rotations, euler parameters $\leq \frac{1}{2}$ ($\sum_{i=1}^4 \epsilon_i^2 = 1$)

from euler parameters to euler angles

$$\begin{aligned}\epsilon_1 &= \sin \frac{\beta}{2} \cos \frac{\alpha - \gamma}{2} & \epsilon_3 &= \cos \frac{\beta}{2} \sin \frac{\alpha + \gamma}{2} \\ \epsilon_2 &= \sin \frac{\beta}{2} \sin \frac{\alpha - \gamma}{2} & \epsilon_4 &= \cos \frac{\beta}{2} \cos \frac{\alpha + \gamma}{2}\end{aligned}$$

Trajectory: position, velocity, acceleration



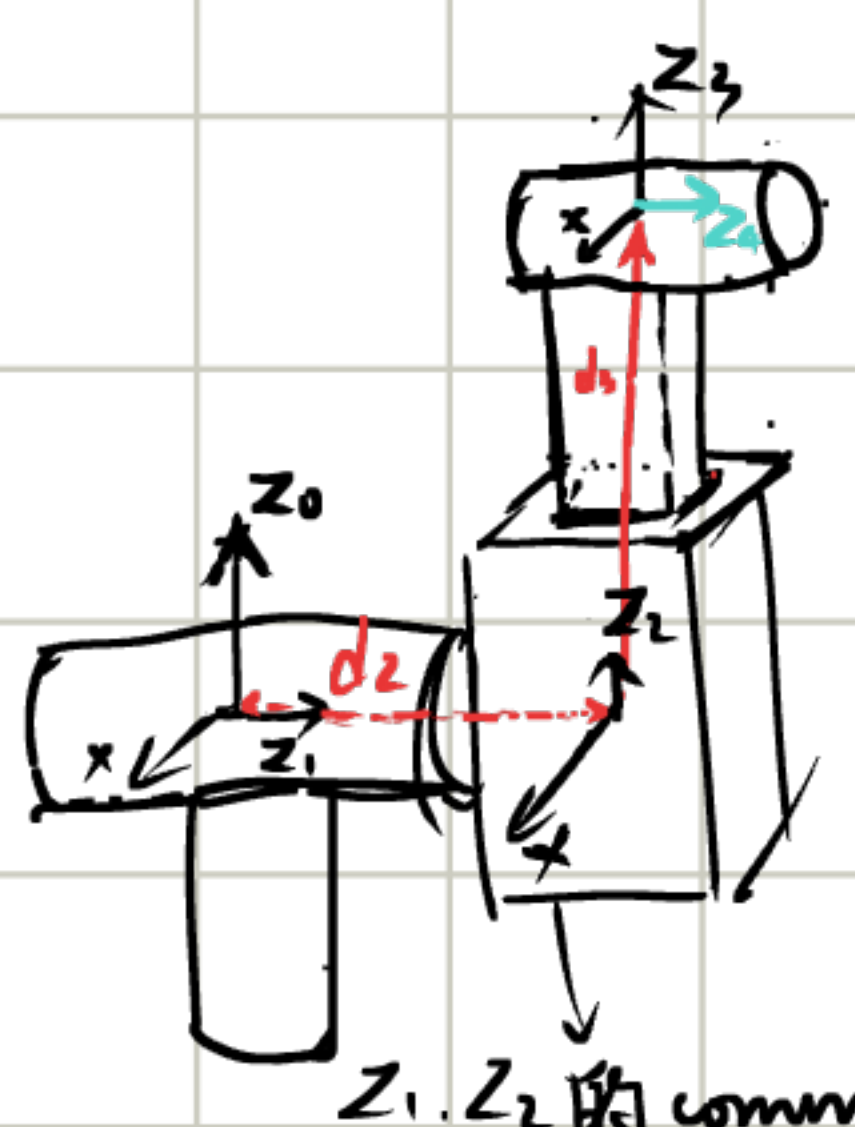
Link Type > Revolute, Prismatic, Spherical, Screw → 螺丝

D-H model (a, d, θ, α) 用于描述 Link.

a_i : link length d_i : distance between two normals

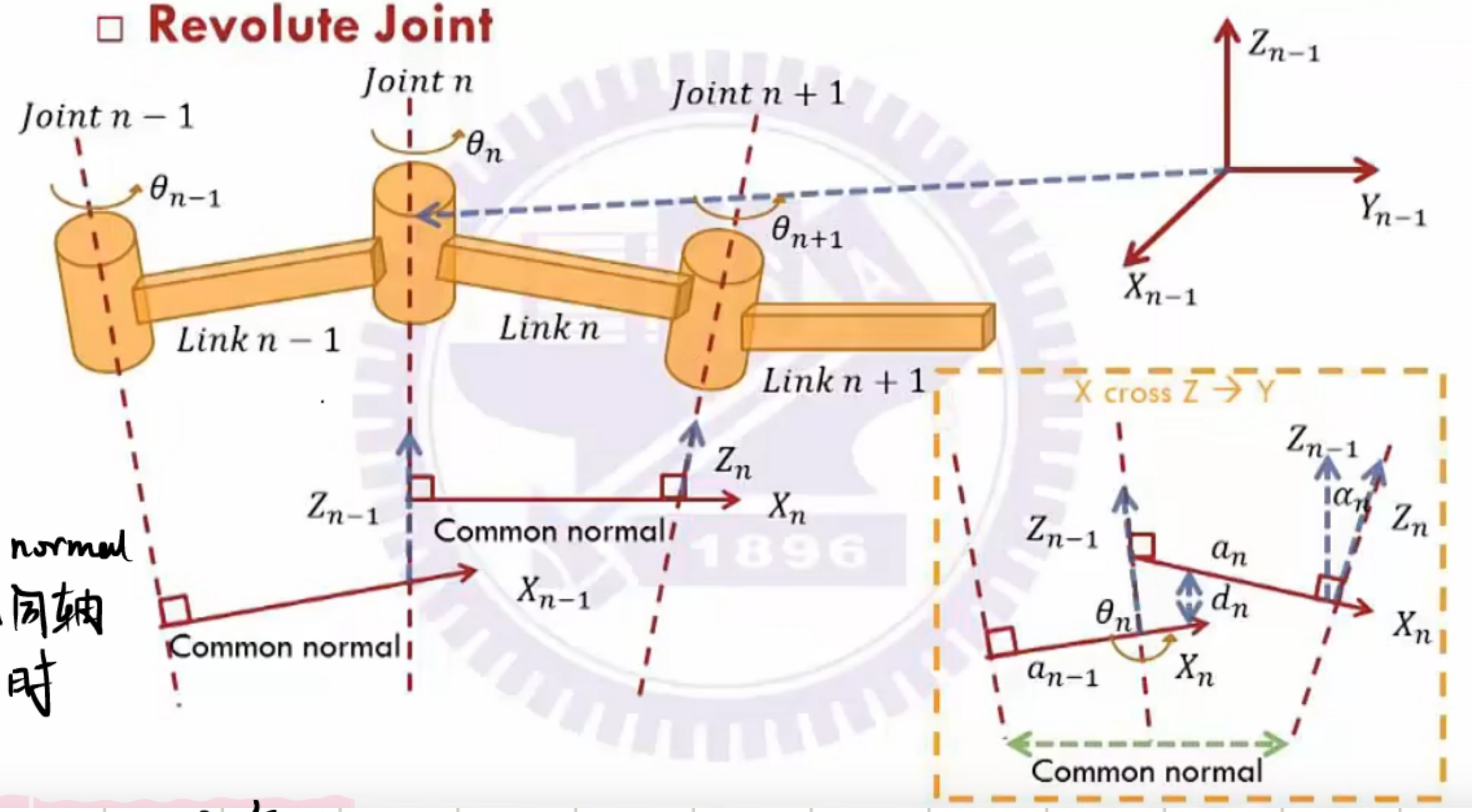
θ_i : angle between two links α_i : link twist

D-H model 要求 link.n, 头部旋转轴为 Joint n 末端为 Z_n



Z_1, Z_2 的 common normal
由于机器人实际设计时不同轴之间的角度为 90° , 所以分析时更加简单. (便于分析 IK)

Revolute Joint



对于旋转轴: θ_i 是变量

① 绕 Z_{n-1} 旋转 θ_n . ② translate along Z_{n-1} a distance d_n

③ 得到新的 X 轴, 沿 X_n 前进 a_n , ④ 绕 X_n 旋转 α_n

对于平行轴: 不考虑平行轴 Joint, Z_n 为下一个旋转轴

$$A_n = \text{Rot}(Z, \theta_n) \text{Trans}(0, 0, d_n) \text{Trans}(a_n, 0, 0) \text{Rot}(X, \alpha_n)$$

$$= \begin{pmatrix} c\theta_n & -s\theta_n c\alpha_n & s\theta_n s\alpha_n & a_n c\theta_n \\ s\theta_n & c\theta_n c\alpha_n & -c\theta_n s\alpha_n & a_n s\theta_n \\ 0 & s\alpha_n & c\alpha_n & d_n \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\text{POS} = \text{BASE} \cdot T_N \cdot \text{Tool}$$

$$T_N = A_1 A_2 \dots A_n$$

Stanford 课程中, 定义的顺序不同, A_n 目的是定义机器人 joint 结构, 强调 θ 的传递性.

$${}^{i-1}T_{(a_{i-1}, a_i, \theta_i, d_i)} = R_x(\alpha_{i-1}) D_x(a_{i-1}) R_z(\theta_i) D_z(d_i)$$

$${}^{i-1}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^0T = {}^0_1T$$

在 prismatic joint 中 d_n 为变量, $a_n = 0$, θ_n 为定值, α_n 通常为 0 或 90°

d_n 开始为 0 ,

每个机器人有一个 Kinematics Table

Inverse Kinematics

$\begin{cases} \text{Base frame;} & \text{Tool frame;} & \text{Goal frame} \\ \text{wrist frame;} & \text{station frame;} & \text{Camera frame.} \end{cases}$

Given T_N , solve θ_1 to θ_N (一个非线性的问题)

线性系统通常有 1 个平衡点, IK 由于有 $\cos\theta, \sin\theta$ 一定不是线性系统.

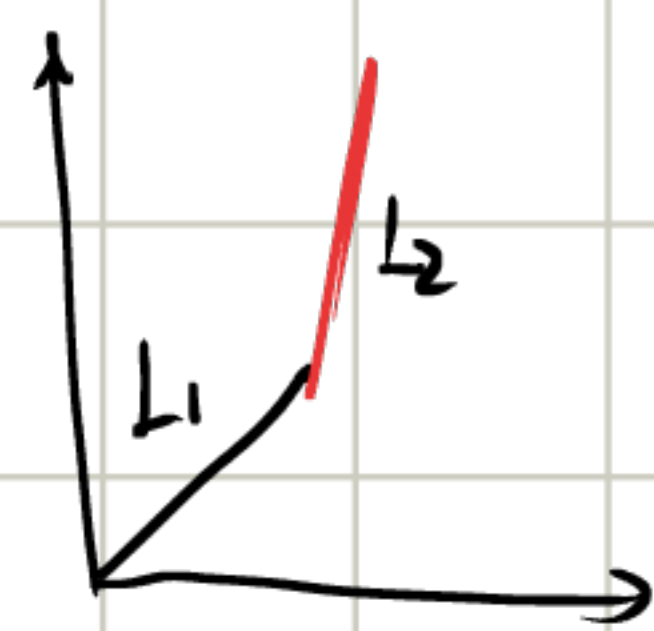
分析非线性系统的方法: 在小区间看作线性系统

如果自由度匹配, 用“数值方法”; 对多数 manipulator 通常没有闭式解

wrist position 的 6 轴 机器人有 close form solution

workspace: end-effector can reach

dexterous workspace: reachable workspace (at least one direction)



当 $L_1 = L_2$ 时, 只有原点是 dexterous workspace.

当 $L_1 \neq L_2$ 时, 没有 dexterous workspace

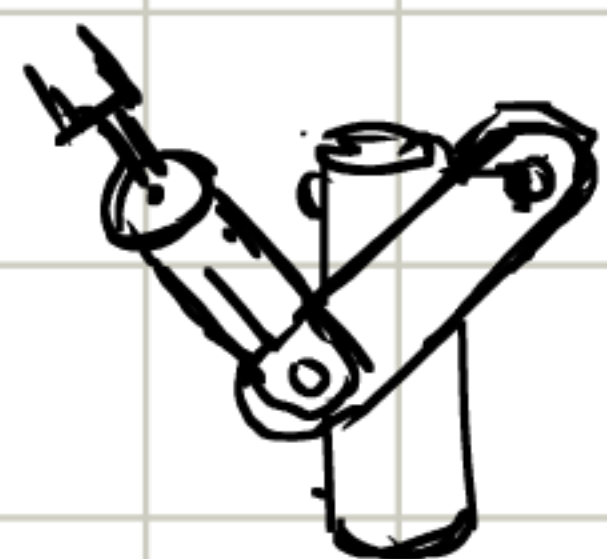
A position may be reached with different configuration

自由度为6. 不代表一对一, 一个 position 有可能对应多个 configuration.

代数法: 对于 PUMA, 写出 A_1, A_2, A_3 的代数表示, 乘起来

等于 T_3 ; 由于只有3个自由度, 一定要先用 position 求解.

首先表示出 A_1, A_2, A_3



解出 θ_1 后看 2, 3 轴

$$T_1 = T_1, T_3 = A_1 T_3, \Rightarrow A_1^{-1} T_3 = T_3 = A_2 A_3$$

最后检查 $\theta_1, \theta_2, \theta_3$ 是否在 joint range 里.

几何法要根据几何关系判断是否有两组解。

对于 6 轴机器人:

① 使用 wrist position 与 rotation 解耦, 分别考虑 3 个自由度.

如果 Tool 没有长度是可行的, 但如果 tool 有长度不能折开

② 通过 Tool 的位置和方向推出 wrist position. 然后使用前 3 轴

$\begin{cases} \text{Righty and above} \\ \text{Lefty and below} \end{cases}$

$\begin{cases} \text{Righty and below} \\ \text{Lefty and above} \end{cases}$

重复性和精确度

Accuracy is bounded by repeatability.

给机器人的小误差做矫正: ① 几何误差 ② 温度, 材料

D-H model 有 4 个参数, 用于运动学学和逆向运动学可以处理

但是不能用四个参数矫正 6 个误差, 需要引入参数 β

达到目标点有 inverse kinematics 和测量的关节角, 所以目标点越多越好

第二部分 Jacobian

$$\delta x_{(m \times 1)} = J_{(m \times n)}(q) \delta q$$

微分关系

给定卡氏坐标系速度, 求轴坐标系速度;

运动产生微分量, 除了速度, 也可以看作微小变化量

Linear velocity: 向量表示.

$$\underline{A}V_Q = \underline{A}V_{Bmg} + \underline{A}R^B V_Q \quad (\text{速度相对于不同坐标系的转换})$$

Rotational velocity: 旋转运动带来直线运动

例如 ${}^B Q$ 是在 frame B 中固定的旋转速度, 而 ${}^A \Omega_B$ 是 {B} 相对于 A 的旋转速度, 需要求出 Q 在 {A} 下的 rotational velocity

一个简单的例子:

$$\text{有 } |\Delta Q| = (|Q| \sin \theta) (|\Omega_B \Delta t|), \quad \underline{A}V_Q = \underline{A}\Omega_B \times \underline{A}Q$$

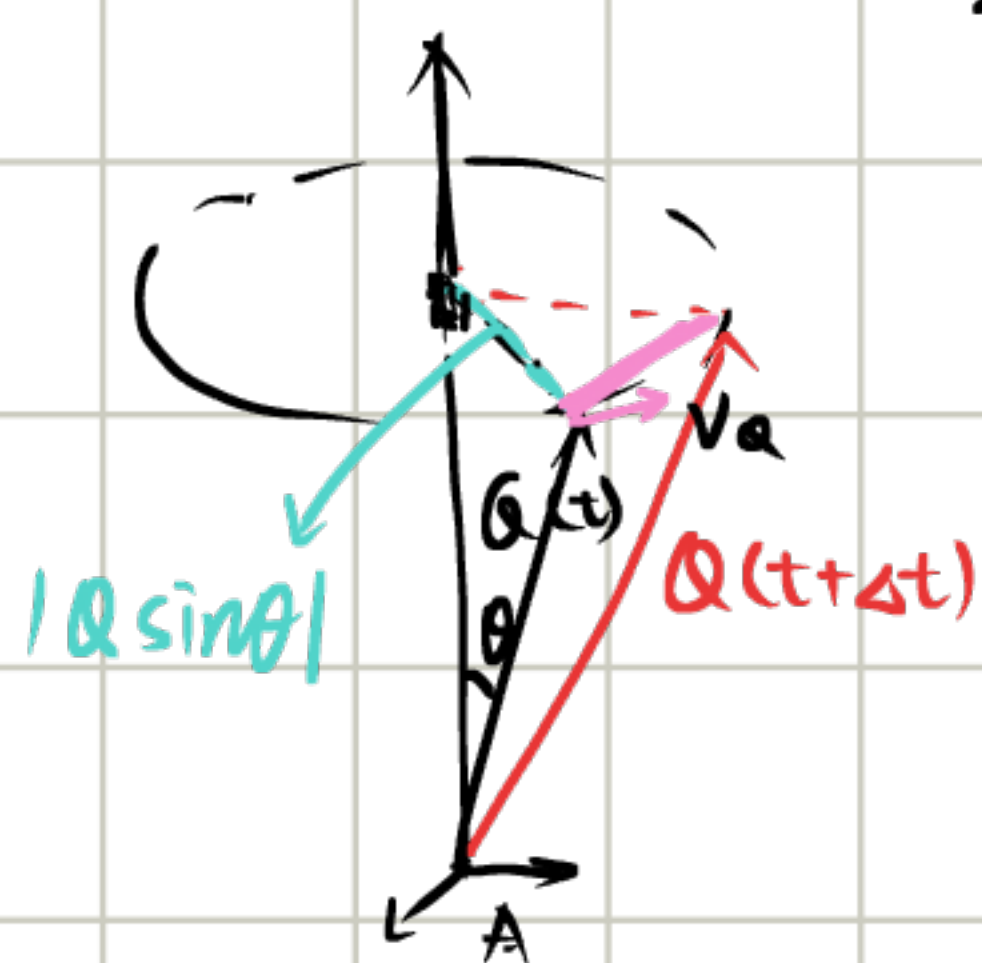
当 Q 在 B 下改变时 (更一般)

$$V_{Q/A} = V_{B/A} + V_{Q/B} + \Omega \times P_B$$

$$\underline{A}V_Q = \underline{A}({}^B V_Q) + \underline{A}\Omega_B \times \underline{A}Q$$

$$= \underline{A}V_B + \underline{A}R_B {}^B V_Q + \underline{A}\Omega_B \times (\underline{A}R_B {}^B Q)$$

粉色为直线速度. $\underline{A}V_Q$



所以 Joint i 对 Joint j 的 linear velocity 贡献为 ($i < j$)

$$v = \Omega_i P_{ij}$$

关键是求 J 的逆和讨论 J 不可逆的情况

常用的参考坐标系 Base / Tool / wrist

$$dx = J dq$$

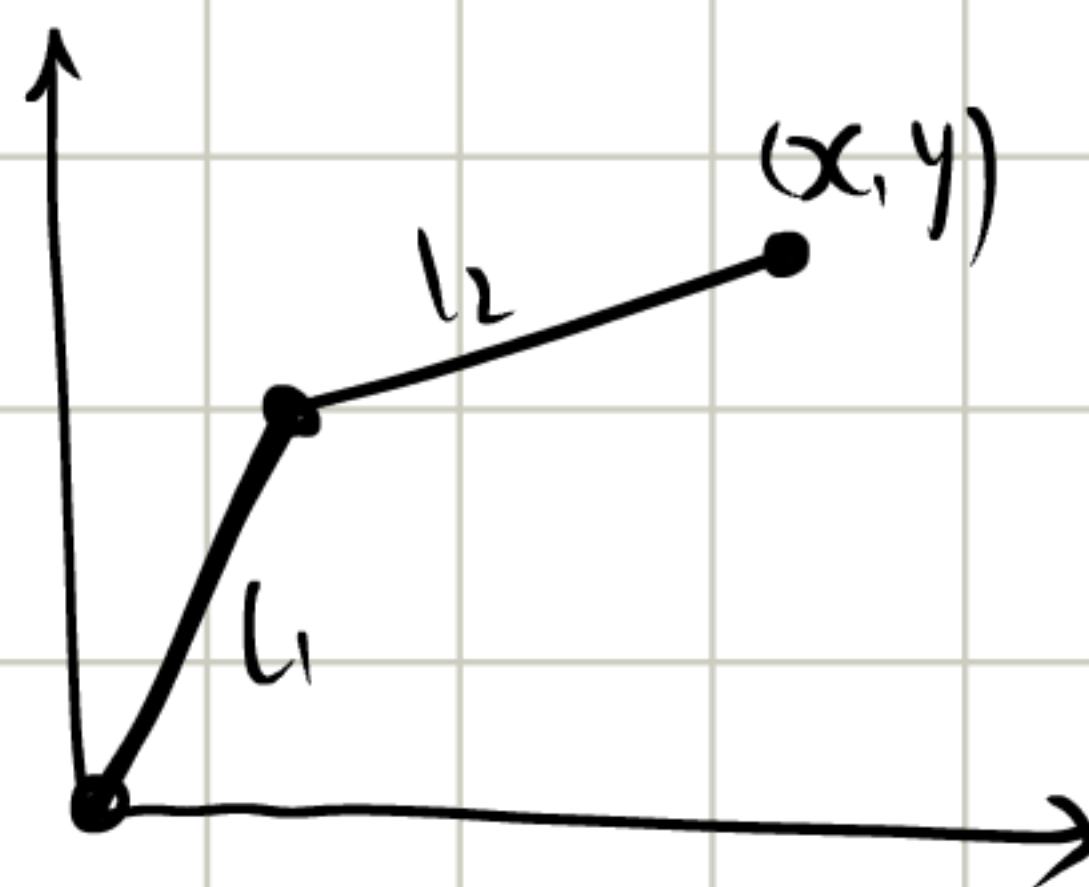
$$dx = \begin{bmatrix} dx \\ dy \\ dz \\ \delta x \\ \delta y \\ \delta z \end{bmatrix}$$

$$dq = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_n \end{bmatrix}$$

$$\frac{dx}{dt} = \frac{df(q)}{dt} = \boxed{\frac{df(q)}{dq}} \times \frac{dq}{dt}$$

$\dot{x} \qquad \qquad \qquad J \qquad \qquad \qquad \dot{q}$

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$



根据 H-D model 有 $x = \begin{bmatrix} x_p \\ r_1 \\ r_2 \\ r_3 \end{bmatrix} = f(\{\alpha_i, a_i, d_i, \theta_i \mid i=1, \dots, 6\})$
 $\in \mathbb{R}^9$

根据表达式求导可以求出该 robot 对应的 Jacobian

$$\dot{x}_{p(3 \times 1)} = J_{x_p(3 \times 6)}(q) \dot{q}_{(6 \times 1)}, \quad x_R \text{ 为 eef Rotation matrix 的列向量}$$

$$x_R = \begin{bmatrix} r_1(q) \\ r_2(q) \\ r_3(q) \end{bmatrix}$$

Basic Jacobian J_0 (对 v 与 ω 的一般表示)

$$\begin{pmatrix} v \\ \omega \end{pmatrix} = J_0(q) \cdot \dot{q}, \quad J_x(q) = \begin{bmatrix} E_p & 0 \\ 0 & E_r \end{bmatrix} \begin{bmatrix} J_v \\ J_\omega \end{bmatrix} = E(x) J_0(q)$$

其中 J_0 是对 v 和 ω 的微分关系,

J_x 是对特定的姿态表示参数的导数和关节速度关系, $E(x)$ 是关联在 Cartesian Coordinates (x, y, z) 中: $E_p = I_3$,

在 Cylindrical Coordinates 中 $(x, y, z) = (p \cos \theta, p \sin \theta, z)$

同理有球坐标系表示,

对于 Euler Angles $x_R = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$,

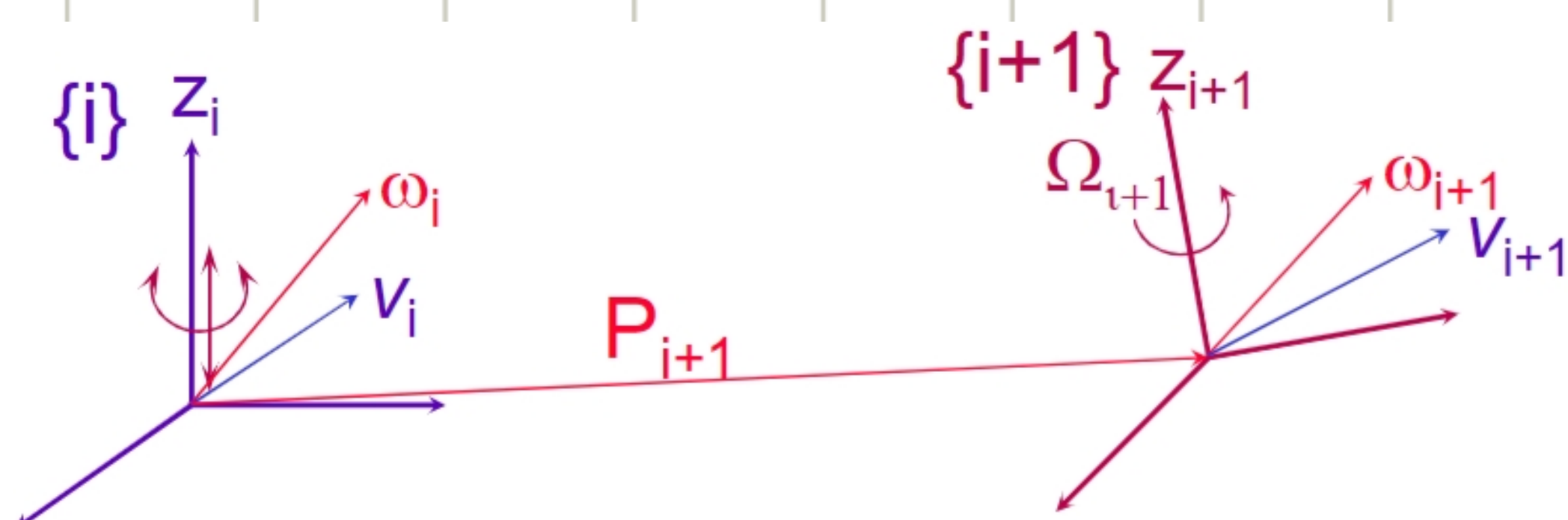
$$E_R(x_R) = \begin{bmatrix} -\frac{s\alpha \cdot c\beta}{s\beta} & \frac{c\alpha \cdot c\beta}{s\beta} & 1 \\ c\alpha & s\alpha & 0 \\ \frac{s\alpha}{s\beta} & -\frac{c\alpha}{s\beta} & 0 \end{bmatrix}$$

在线速度和角速度的视角下求解 J_0 .

velocity propagation 速度在两个关节坐标系中的传递

$$\text{Linear: } v_{i+1} = v_i + \omega_i \times p_{i+1} + \dot{d}_{i+1} z_{i+1}$$

$$\text{Angular: } \omega_{i+1} = \omega_i + \Omega_{i+1}; \quad \Omega_{i+1} = \dot{\theta}_{i+1} z_{i+1}$$



首先求出在 $\{n\}$ 坐标系下的关节 n 的线速度和角速度

对机器人: Angular velocity of link $i+1$, ω_{i+1} (角速度的传递)

Joint 1: v_1 and ω_1 → 在 $\{i+1\}$ 下看 i

Joint $i+1$: ${}^{i+1}\omega_{i+1} = \underbrace{{}^iR \cdot {}^i\omega_i}_{\text{角速度的传递}} + \dot{\theta}_{i+1} \cdot {}^{i+1}Z_{i+1}$

${}^{i+1}v_{i+1} = \underbrace{{}^iR \cdot ({}^iv_i + {}^i\omega_i \times {}^iP_{i+1})}_{\text{速度的传递}} + \dot{d}_{i+1} \cdot {}^{i+1}Z_{i+1}$

$$\begin{pmatrix} {}^0v_n \\ {}^0\omega_n \end{pmatrix} = \begin{bmatrix} {}^0R & 0 \\ 0 & {}^0R \end{bmatrix} \begin{pmatrix} {}^nv_n \\ {}^n\omega_n \end{pmatrix} \quad \left\{ \begin{array}{ll} V_i = Z_i \dot{\theta}_i & \text{Prismatic } \in \\ \Omega_i = Z_i \dot{\theta}_i & \text{Revolute } \bar{\in} \end{array} \right.$$

End effector velocity

Jacobian explicit form

Linear velocity: $v = \sum_{i=1}^n [E_i V_i + \bar{E}_i (\Omega_i \times P_{in})]$

Angular velocity: $\omega = \sum_{i=1}^n \bar{E}_i \Omega_i$

$v = \sum_{i=1}^n [E_i Z_i + \bar{E}_i Z_i \times P_{in}] \dot{\theta}_i$

$= [E_1 Z_1 + \bar{E}_1 Z_1 P_{in}, \dots] \begin{bmatrix} \dot{\theta}_1 \\ \vdots \\ \dot{\theta}_n \end{bmatrix} = J_v \dot{\theta}$

$\omega = [\bar{E}_1 Z_1, \bar{E}_2 Z_2, \dots, \bar{E}_n Z_n] \begin{bmatrix} \dot{\theta}_1 \\ \vdots \\ \dot{\theta}_n \end{bmatrix} = J_\omega \dot{\theta}$

因此, 实际求解 Jacobian 时
可以直接利用各个 Joint 的状态
得到

于是有

同理, 在 frame $\{0\}$ 下的 Jacobian 为

$J = \begin{pmatrix} \frac{\partial x_p}{\partial \theta_1} & \dots & \frac{\partial x_p}{\partial \theta_n} \\ \bar{E}_1 Z_1 & \dots & \bar{E}_n Z_n \end{pmatrix}$

${}^0J = \begin{pmatrix} \frac{\partial ({}^0x_p)}{\partial \theta_1} & \dots & \frac{\partial ({}^0x_p)}{\partial \theta_n} \\ \bar{E}_1 ({}^0R \cdot Z), \dots, \bar{E}_n ({}^0R \cdot Z) \end{pmatrix}$

其中 $Z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$; 进一步 ${}^B J = \begin{bmatrix} {}^B_R & 0 \\ 0 & {}^B_{AR} \end{bmatrix} {}^A J$

Kinematic Singularity

$\det(J) = 0$, 有性质 $\det({}^B J) = \det({}^A J)$

对 EEF 有 $\begin{cases} v_e = v_n - P_{ne} \times \omega_n \\ \omega_e = \omega_n \end{cases}$

于是有 ${}^0J_e = \begin{bmatrix} I & -{}^0\hat{P}_{ne} \\ 0 & I \end{bmatrix} {}^0J_n$, ${}^0\hat{P} = {}^0R^n \hat{P}_n {}^0R^T$

$${}^0J_e = \begin{bmatrix} {}^0R & -{}^0R \hat{p}_{ne} {}^0R^T \\ 0 & {}^0R \end{bmatrix} {}^nJ_n$$

另一个版本的 Jacobian 推导: 关注 pose 的变化. (没太清楚)
T 为某个 joint 的 pose.

$$T + dT = \text{Trans}(dx, dy, dz) * \text{Rot}(k, d\theta) * T \quad (\text{对于 Axis-angle})$$

$$T_{\Delta} = \begin{bmatrix} 0 & -dz & dy & dx \\ dz & 0 & -dx & dy \\ -dy & dx & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{做轻微扰动}), \text{对某一个特定 joint 做扰动}$$

$$[A_i + {}^{i-1}\Delta_i \cdot A_i \cdot d\theta_i]$$

$$\text{对 Joint } N \text{ 影响: } T_N + dT_N = A_1 * A_2 * \dots * (A_i + dA_i) * \dots * A_N$$

$$T_N + dT_N = T_N + (A_1 * \dots * A_{i-1} * {}^{i-1}\Delta_i * A_i * A_{i+1} * \dots * A_N) d\theta_i$$

$$\text{可以得到: } dT_N = T_N * {}^N\Delta_i * d\theta_i, \text{接下来只需关注 } {}^N\Delta_i$$

$$\text{分类讨论 Joint } i. \quad \textcircled{1} \text{ 对于 revolute joint } {}^{i-1}d_i = 0, {}^{i-1}s_i = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$${}^N\Delta_i = \begin{bmatrix} n \cdot [(^{i-1}s_i \times p) + {}^{i-1}d_i] \\ 0 \cdot [(^{i-1}s_i \times p) + {}^{i-1}d_i] \\ a \cdot [(^{i-1}s_i \times p) + {}^{i-1}d_i] \end{bmatrix} = \begin{bmatrix} p_x n_y - n_x p_y \\ p_x o_y - o_x p_y \\ p_x a_y - a_x p_y \end{bmatrix}$$

$$\text{其中的 } n, o, a, p \text{ 对应 } U_{i-1} = A_i * A_{i+1} * \dots * A_N$$

$${}^N s_i = \begin{bmatrix} n \cdot {}^{i-1}s_i \\ o \cdot {}^{i-1}s_i \\ a \cdot {}^{i-1}s_i \end{bmatrix} = \begin{bmatrix} n_z \\ o_z \\ a_z \end{bmatrix}$$

${}^N d_i$ 为第 i 个对末端坐标系 $\{N\}$ 中原点位置平移贡献

$$\textcircled{2} \text{ prismatic joint: } {}^{i-1}d_i = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, {}^{i-1}s_i = 0, {}^N d_i = \begin{bmatrix} n_z \\ o_z \\ a_z \end{bmatrix}, {}^N s_i = 0$$

$$\text{综上 } {}^N\Delta = \sum_{i=1}^N (U_{i-1}^{-1} * {}^{i-1}\Delta_i * U_{i-1}) \quad {}^N\Delta \text{ 为累加到末端的微分变化量}$$

$$\begin{bmatrix} {}^N d \\ {}^N s \end{bmatrix} = \begin{bmatrix} {}^N\Delta_1 & {}^N\Delta_2 & \dots & {}^N\Delta_N \end{bmatrix} \begin{bmatrix} d\theta_1 \\ \vdots \\ d\theta_N \end{bmatrix}$$

Joint 1 对 ${}^N\Delta$ 的贡献

J

Inverse Jacobian

① R. Paul's method

a). from T_6 & $T_6 S$ compute $dT_6 = T_6 * T_6 \Delta = \begin{pmatrix} dn_6 & do_6 & da_6 & dp_6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

b) 使用 Kinematic equation 求微分, 用 $d\theta$'s 与 dT_6 的关系.
未直接求 J^{-1}

② Used in the industries in the past → new T

a) compute dT_6 , b) compute $T_6^N = T_6 + dT_6 \Rightarrow T^T \Delta = \frac{\partial T \Delta}{\partial q} \partial q$

c) compute new joint solution from T_6^N ,

d) $dq = q^N - q$ (思想: 分别求解两个 T 的 inverse kinematics)

③ 神经网络/模糊系统

静力分析 Static Force

对一轴刚体: $v = J\dot{\theta}$, 有 $\tau = J^T F$ (力矩)

对一个刚体, 由 Static Equilibrium: $\sum \text{Forces} = 0$

对 prismatic joint $\tau_i = f_i^T Z_i$, 对 revolute joint $\tau_i = n_i^T Z_i$

$$\begin{cases} {}^n f_n = {}^n f \\ {}^n n_n = {}^n n + {}^n P_{n+1} \times {}^n f \end{cases} \quad \text{坐标转换}$$

$$\begin{cases} {}^i f_i = {}^i R^{i+1} {}^i f_{i+1} \\ {}^i n_i = {}^i R^{i+1} {}^i n_{i+1} + {}^i P_{i+1} \times {}^i f_i \end{cases}$$

其中有 $F = \begin{pmatrix} f \\ n \end{pmatrix}$ 定义为 Wrench

n 为力矩向量, 表示该点的旋转力矩

虚功原理, $\tau = J^T \begin{pmatrix} f \\ n \end{pmatrix}$, τ : 关节驱动器需要提供用力矩

第三部分: Perception and Sensing

Non-Verged Binocular Stereo System. 假设 y 相同

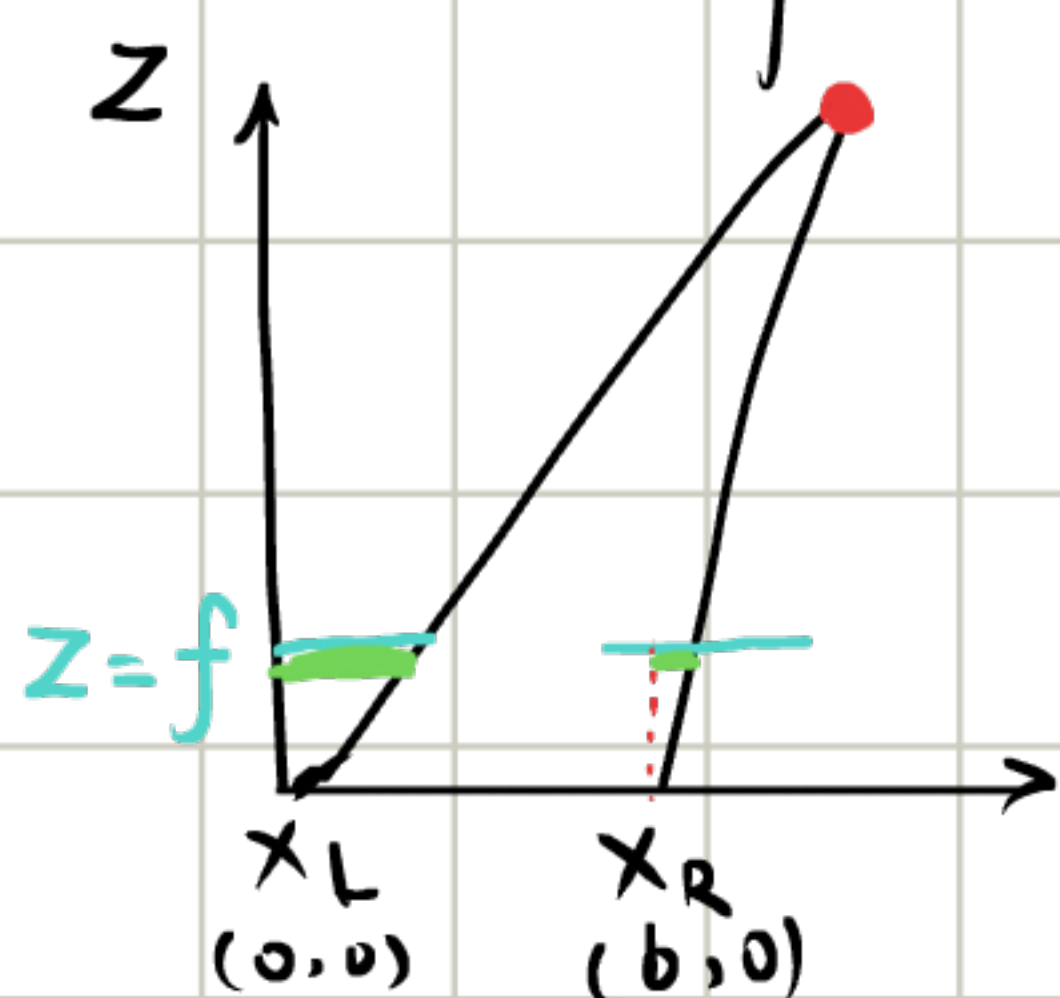
有 S_x, S_y 分别为水平方向的焦距

$$\text{于是 } X_L = S_x \frac{X}{Z}$$

$$X_R = S_x \frac{X-b}{Z}$$

$$Y_L = Y_R = S_y \frac{Y}{Z}$$

X_R, X_L, Y_L, Y_R 为左右图片的像素坐标



视差 $D = (X_L - X_R)$, $Z = \frac{b S_x}{D}$ 得到 Z 的表达式

两个相机中三维点的关系：设左侧相机中为 P_L ，右侧为 P_R

$P_R = R(P_L - T)$ ，由共面约束：点 P, C_L, C_R (相机光心)

$$P_R^T (T \times P_L) = 0 \Rightarrow P_R^T [T] P_L = 0$$

$\Rightarrow$ 定义 $P_L = R^T P_R$ ，拉到右相机坐标系中

$$P_R^T [T] \times R P_L = 0$$

定义 foundation matrix $E = [T] \times R$

Matching and Correspondence

{ feature based
region based

第四部分 Trajectory Planning

考虑避开障碍物

障碍物：{ 会动
 不会动 } { 可预期
 不可预期
 可控制
 不可控制

设置路径时位置连续、速度连续，加速度可以不连续

避开 single point.

path description: 起点、终点和路径中的一些点.

使用 Cartesian 坐标系面临的问题:

① 起点和终点可达，但是有的中间点不可达

② 在奇点处速度达到 ∞ ，导致不可解

③ 到达过程有多解

1. 使用 Cubic Polynomial 去拟合 translation 或 rotation 的变量，如果

初始和结束速度为 0，三次多项式可以平滑速度和加速度

2. 使用线性，但可以在开始运动和截止运动时逐步改变加速度进行平滑

3. 也可使用高阶多项式拟合 $u(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4$

第五部分: 动力学 Dynamics

$$M(q) \ddot{q} + V(q, \dot{q}) + G(q) = P$$

q : Joint Coordinate; $M(q)$: Mass matrix

$V(q, \dot{q})$: Centrifugal and Coriolis forces

$G(q)$: Gravity forces

P : Generalized forces

方法 ① Newton-Euler

② Lagrange

Newton-Euler

$$F = ma \Rightarrow m\dot{v} = F \quad \text{定义 } \phi = mv$$

推广到角动量 $p \times m\dot{v} = p \times F$

对旋转运动 $v_i = \omega \times p_i$ (对质点 i 的线速度)

$$\begin{aligned} \phi &= \sum_i m_i p_i \times (\omega \times p_i), \quad m_i = \rho dv \\ &= \int_V p \times (\omega \times p) \rho dv \end{aligned}$$

$$p \times (\omega \times p) = \hat{p}(-\hat{p})\omega \Rightarrow \phi = \left[\int_V -\hat{p}\hat{p} \rho dv \right] \omega = I\omega$$

Inertia Tensor

于是有 $\begin{cases} \dot{\phi} = F & \text{Linear Momentum} \\ \dot{\phi} = N & \text{Angular Momentum} \end{cases}$

parallel Axis Theorem: (惯量矩阵在不同坐标系间关系)

$$\text{有 } I_A = I_c + m[(p_c^T p_c)I_3 - p_c p_c^T]$$

由于 $\begin{cases} m\dot{v}_c = F \\ I_c \dot{\omega} + \omega \times I_c \omega = N \end{cases}$ (由 $\phi = I\omega$ 求导)

角加速度: 已知第 i 个坐标系的角速度/角加速度得到第 $i+1$ 个坐标系

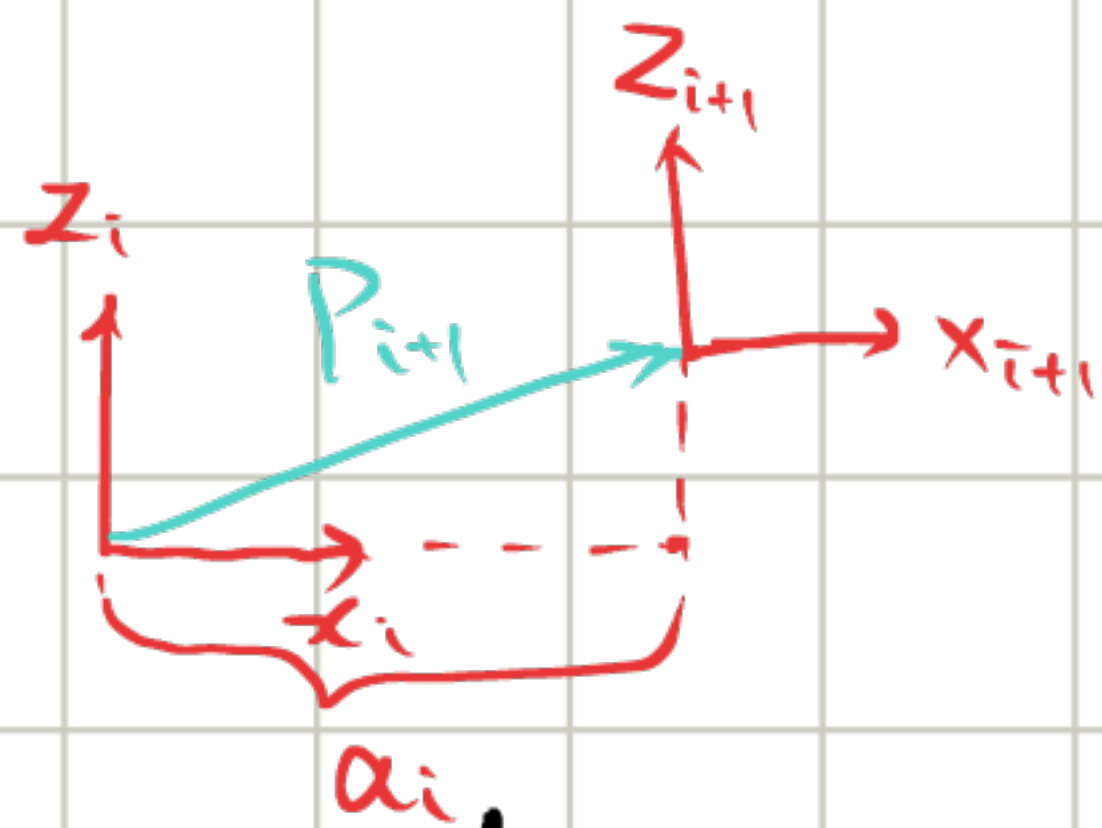
(ω_i 第 i 个坐标系相对于基座)

$$\begin{aligned} \omega_{i+1} &= \omega_i + \Omega_{i+1} \\ \Omega_{i+1} &= \dot{\theta}_{i+1} Z_{i+1} \Rightarrow \dot{\omega}_{i+1} = \dot{\omega}_i + \dot{\Omega}_{i+1} = \dot{\omega}_i + \dot{\theta}_{i+1} Z_{i+1} + \dot{\theta}_{i+1} Z_{i+1} \end{aligned}$$

由于对 V 随刚体转动的单位向量 a , 有 $\dot{a} = \omega \times a$.

$$\dot{\omega}_{i+1} = \dot{\omega}_i + \dot{\theta}_{i+1} (\omega_i \times Z_{i+1}) + \ddot{\theta}_{i+1} Z_{i+1}$$

线加速度 $v_{i+1} = v_i + w_i \times p_{i+1} + \dot{d}_{i+1} z_{i+1}$



$$P_{i+1} = a_i x_i + d_{i+1} Z_{i+1}$$

于是有 $\dot{v}_{i+1} = \dot{v}_i + \dot{\omega}_i \times p_{i+1} + \omega_i \times (\omega_i \times p_{i+1}) + 2\overset{u}{\dot{d}_{i+1}} \omega_i \times z_{i+1} + \ddot{d}_{i+1} z_{i+1}$

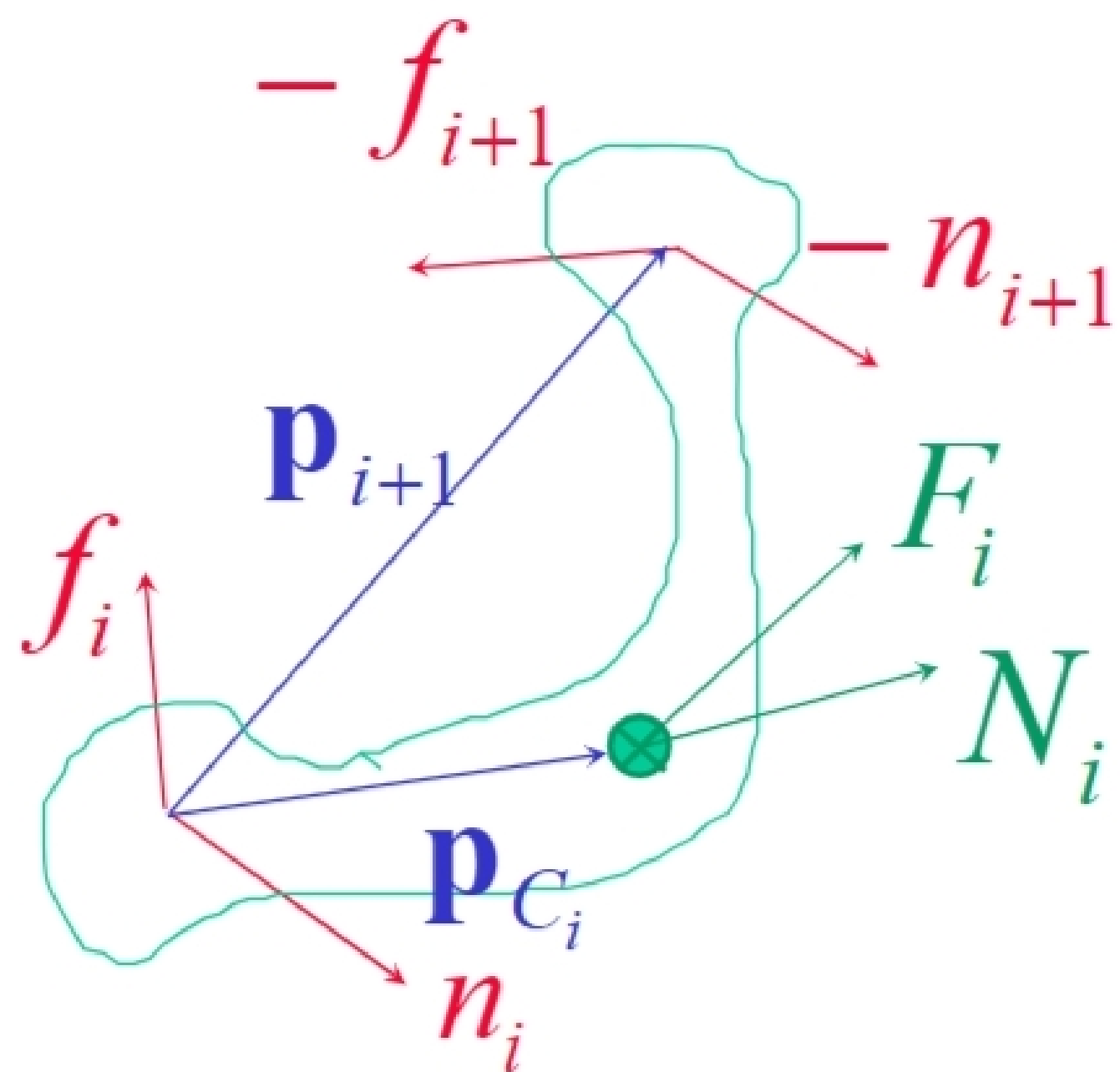
刚体质心处的速度及加速度.

$$v_{C_{i+1}} = v_{i+1} + \omega_{i+1} \times p_{C_{i+1}},$$

$$\dot{V}_{C_{i+1}} = \dot{V}_{i+1} + \dot{\omega}_{i+1} \times P_{C_{i+1}} + \omega_{i+1} \times (\omega_{i+1} \times P_{C_{i+1}})$$

对某个 Link: $F_i = f_i - f_{i+1}$

$$N_i = n_i - n_{i+1} + (-P_{C_i}) \times f_i + (P_{i+1} - P_{C_i}) \times (-f_{i+1})$$



Recursive Equations

$$f_i = F_i + f_{i+1}$$

$$n_i = N_i + n_{i+1} + P_{c_i} \times F_i + P_{i+1} \times f_{i+1}$$

$$Z_i = \begin{cases} n_i Z_i & \text{revolute} \\ f_i Z_i & \text{prismatic} \end{cases}$$

其中 $F_i = m_i \dot{v}_{Ci}$, $N_i = I_{Ci} \dot{\omega}_i + \omega_i \times I_{Ci} \omega_i$

实际应用时,可根据运动学 $i = 0, \rightarrow, 5$, 求出

$$\{ {}^{i+1}w_{i+1}, {}^{i+1}\dot{w}_{i+1}, {}^{i+1}v_{i+1}, {}^{i+1}\dot{v}_{i+1}, {}^{i+1}F_{i+1}, {}^{i+1}N_{i+1} \} \quad (\text{Outward iteration})$$

Inward iteration. $i: 6 \rightarrow 1$.

$$\vec{f}_i = {}_{i+1}R^i \vec{f}_{i+1} + \vec{F}_i$$

Lagrange Equation

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = \tau.$$

Lagrangian $L = K - U \rightarrow$ Potential Energy
 Kinetic Energy $\rightarrow$ 动能变化 $\rightarrow$ 位置变化.

对于 $U = U(q)$, $\frac{d}{dt} \left(\frac{\partial K}{\partial \dot{q}} \right) - \frac{\partial K}{\partial q} + \frac{\partial U}{\partial q} = \tau$.
 inertial force Gravity vector G

记 $K = \frac{1}{2} \dot{q}^T M(q) \dot{q}$, 有 $\frac{\partial K}{\partial \dot{q}} = M(q) \dot{q}$
 $\frac{d}{dt} (M(q) \dot{q}) - \frac{\partial K}{\partial q} = M \ddot{q} + \dot{M} \dot{q} - \frac{1}{2} \dot{q}^T \frac{\partial M}{\partial q} \dot{q} = M \ddot{q} + V(q, \dot{q})$
 于是有 $M(q) \ddot{q} + V(q, \dot{q}) + G(q) = \tau$

机械 Kinetic Energy

$K = \frac{1}{2} m v^2$ (对 v). $K = \frac{1}{2} \omega^T I_c \omega$ (对 ω)

则一般的 $K_i = \frac{1}{2} (m_i v_{ci}^T v_{ci} + \omega_i^T I_{ci} \omega_i)$

总动能 $K = \sum_i K_i$

已知 $\begin{cases} v_{ci} = J_{v_i} \dot{q} \\ \omega_{ci} = J_{\omega_i} \dot{q} \end{cases}$ 则 $M = \sum_{i=1}^n (m_i J_{v_i}^T J_{v_i} + J_{\omega_i}^T I_{ci} J_{\omega_i})$
 (观察到 $M^T = M$)

Potential Energy

$U_i = m_i g \cdot h_i + U_0 = m_i (-g^T p_{ci})$

$G_j = \frac{\partial U}{\partial q_j} = - \sum_{i=1}^n (m_i g^T \frac{\partial p_{ci}}{\partial q_j})$

$G = - \begin{pmatrix} J_{v_1}^T & J_{v_2}^T & \dots & J_{v_n}^T \end{pmatrix} \begin{pmatrix} m_1 g \\ \vdots \\ m_n g \end{pmatrix}$
 (涉及到) Jacobian matrix

$\tau = r \times F$ (τ : torque / moment)

$\tau_j = z_j^T (r \times F)$ (对特定关节)

从电机角度: $\tau_{motor} = K_t i$, K_t 为转矩常数, i 为电流

$\tau_{joint} = \eta N \tau_{motor}$, η 效率, N : 减速比

$\tau = M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q)$ τ 机械输入, 其余项为被动响应

此外: $\delta W = \tau^T \delta q \Rightarrow P = \tau^T \dot{q}$ (与功率的关系)

第六部分: Control

- PID Control.
- Joint space control
- Task-Oriented Control
- Force Control

resolved motion rate control.:

$x = f(\theta), \delta x = x_d - x, \delta \theta = J^{-1} \delta x, \theta^+ = \theta + \delta \theta.$

Natural System: 如质点-弹簧系统

$\frac{d}{dt} \left(\frac{\partial (K-V)}{\partial \dot{x}} \right) - \frac{\partial (K-V)}{\partial x} = 0 \quad K = \frac{1}{2} m \dot{x}^2$

$\Rightarrow m \ddot{x} = F = -kx, \quad V = \frac{1}{2} kx^2$

$\Rightarrow m \ddot{x} + kx = 0$

自然频率. natural frequency: $\omega_n = \sqrt{\frac{k}{m}}, \quad \ddot{x} + \omega_n^2 x = 0$

对于 Dissipative System: $f_{friction} = -b\dot{x}$

有 $m \ddot{x} + b\dot{x} + kx = 0$

- Oscillatory damped
- Critically damped
- Over damped

$\ddot{x} + \frac{b}{m} \dot{x} + \omega_n^2 x = 0$
 $\ddot{x} + 2\zeta_n \omega_n \dot{x} + \omega_n^2 x = 0$

定义 $\zeta_n = \frac{b}{2\omega_n m}$ 为 Natural damping ratio.

当 $b = 2\sqrt{km}$ 时 $\zeta_n = 1$, 及满足 Critically damped

将 $\ddot{x} + 2\omega_n \zeta_n \dot{x} + \omega_n^2 x = 0$ 微分方程求解后得到

$x(t) = c e^{-\zeta_n \omega_n t} \cos(\omega_n \sqrt{1-\zeta_n^2} t + \phi)$

passive systems (对于机械臂只考虑动能) $L = K - V_{goal}$, 无阻尼 无外扰 $L=0$

目标势能. $V_{goal} = \frac{1}{2} k_p (x - x_g)^T (x - x_g) \quad \frac{d}{dt} \left(\frac{\partial K}{\partial \dot{x}} \right) - \frac{\partial K}{\partial x} = f \rightarrow$ 维持原有稳定.

$\times f = -\frac{\partial V_{goal}}{\partial x}, \quad \text{Stable} = \frac{d}{dt} \left(\frac{\partial K}{\partial \dot{x}} \right) - \frac{\partial (K - V_{goal})}{\partial x} = 0$

Asymptotic Stability 平衡点 $x = x_g$.

$\frac{d}{dt} \left(\frac{\partial K}{\partial \dot{x}} \right) - \frac{\partial (K - V_{goal})}{\partial x} = F_s, \quad \text{stable when } F_s^T \dot{x} < 0 \text{ for } \dot{x} \neq 0$

$F_s = -k_v \dot{x} \quad k_v > 0.$

Control: $F = -k_p (x - x_{goal}) - F_s$

Proportional-Derivative Control (PD)

$$m\ddot{x} = f = -k_p(x-x_d) - k_v\dot{x} \Rightarrow m\ddot{x} + \overset{\text{velocity gain}}{k_v}\dot{x} + \overset{\text{position gain}}{k_p}(x-x_d) = 0$$

于是 $\ddot{x} + \frac{k_v}{m}\dot{x} + \frac{k_p}{m}(x-x_d) = 0$

$$\ddot{x} + 2\xi\omega\dot{x} + \omega^2(x-x_d) = 0$$

其中 $\begin{cases} \omega = \sqrt{\frac{k_p}{m}} & \text{closed loop frequency} \\ \xi = \frac{k_v}{2\sqrt{k_p m}} & \text{closed loop damping ratio} \end{cases}$

实际中, 给定 ξ 与 ω , 得到系统的 k_p 和 k_v

有时候不考虑质量, 定义 $k'_p = \omega^2$, $k'_v = 2\xi\omega$, 有 $\ddot{x} + k'_v\dot{x} + k'_p(x-x_d) = 0$

Motion Control

一般表示 加上 $b(x, \dot{x})$

$$m\ddot{x} + b(x, \dot{x}) = f \Rightarrow 1. \ddot{x} = f'$$

goal position: $f' = -k'_v\dot{x} - k'_p(x-x_d)$

trajectory tracking: $x_d(t)$; $\dot{x}_d(t)$, and $\ddot{x}_d(t)$

Control $f' = \ddot{x}_d - k'_v(\dot{x} - \dot{x}_d) + k'_p(x - x_d)$

令 $x - x_d = e$, 则系统满足 $\ddot{e} + k'_v\dot{e} + k'_pe = 0$

引入 disturbance rejection: $m\ddot{x} + b(x, \dot{x}) = f + f_{\text{dist}}$

Steady State Error

$$\ddot{e} + k'_v\dot{e} + k'_pe = \frac{f_{\text{dist}}}{m}, \text{ 当 } \dot{e} = \ddot{e} = 0 \text{ 时达到 steady state}$$

此时 $e = x - x_d$, $x = x_d + \frac{f_{\text{dist}}}{k_p}$

PID system $m\ddot{x} + b(x, \dot{x}) = f + f_{\text{dist}}$

control $f = mf' + b(x, \dot{x})$

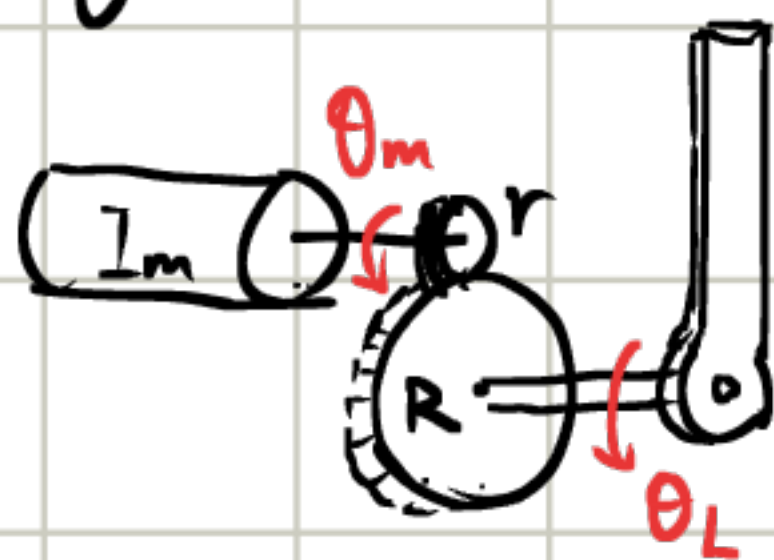
此时 $f' = \ddot{x}_d - k'_v(\dot{x} - \dot{x}_d) - k'_p(x - x_d) - k'_i \int (x - x_d) dt$

$$\Rightarrow \ddot{e} + k'_v\dot{e} + k'_pe + k'_i \int e dt = \frac{f_{\text{dist}}}{m}$$

$$\Rightarrow \ddot{e} + k'_v\dot{e} + k'_pe + k'_ie = 0$$

当 Steady state 时 $e = 0$

Gear Reduction



$$\text{gear ratio} = \eta = \frac{R}{r}$$

$$\theta_L = \left(\frac{1}{\eta}\right)\theta_m; \tau_L = \eta\tau_m$$

根据 $\tau_m = I_m \ddot{\theta}_m + \frac{1}{\eta} (I_L \ddot{\theta}_L) + b_m \dot{\theta}_m + \frac{1}{\eta} b_L \dot{\theta}_L$

$$\Rightarrow \tau_m = \left(I_m + \frac{I_L}{\eta^2}\right) \ddot{\theta}_m + \left(b_m + \frac{b_L}{\eta}\right) \dot{\theta}_m$$

$$\tau_L = (I_L + \eta^2 I_m) \ddot{\theta}_L + (b_L + \eta^2 b_m) \dot{\theta}_L$$

第一项系数为 effective inertia, 第二项为 damping

Effective Inertia

$$I_{\text{eff}} = I_L + \eta^2 I_m \text{ 对于 manipulator } I_L = I_L(q)$$

对于 direct drive $\eta=1$

在选择参数时 $k_p = (I_L + \eta^2 I_m) k_p'$ $k_v = (I_L + \eta^2 I_m) k_v'$

time optional selection $\hat{I}_L = \frac{1}{4} (\sqrt{I_{L_{\min}}} + \sqrt{I_{L_{\max}}})^2$

Task oriented Control. 相比起 Lagrange 使用 x 而不是 q .

$$F = -\nabla V(x_{\text{goal}}), \quad \tau = J^T F$$

对 end effector: $F = F_{\text{motion}} + F_{\text{contact}}$

希望满足 $M_x(x)\ddot{x} + V_x(x, \dot{x}) + G_x(x) = F$

equations for motion: $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = F$ with $L(x, \dot{x}) = K(x, \dot{x}) - U(x)$

$$K_x(x, \dot{x}) = K_q(q, \dot{q}) \Rightarrow \frac{1}{2} \dot{x}^T M_x(x) \dot{x} = \frac{1}{2} \dot{q}^T M(q) \dot{q}$$

代入 $\dot{x} = J \dot{q}$ 得 $M_x(x) = J^{-T}(q) M(q) J^{-1}(q)$

同理 $V_x(x, \dot{x}) = J^{-T}(q) V(q, \dot{q}) - M(q) h(q, \dot{q})$

$$G_x(x) = J^{-T}(q) G(q)$$

where $h(q, \dot{q}) = \dot{J}(q) \dot{q}$

最终, 得到 F 后利用 $\tau = J^T F$ 即可得到实际控制的力矩

Stiffness

$$\ddot{z} + k_v \dot{z} + k_p' (z - z_d) = 0 \quad k_p' \text{ 决定沿 } z \text{ 方向的刚性}$$

k_p' 决定刚度, k_v 决定阻尼.

$$\tau = J^T F = J^T K_x \Delta x = (J^T K_x J) \Delta \theta = K_\theta \Delta \theta$$

其中 $K_\theta = J^T K_x J$

Force Control

由于位控时, 由于环境是刚性的, 接触力较大容易损坏 ($k_e \gg k_d$)

1. 纯力控制, 目标 $F \rightarrow F_d$

$$F_{\text{control}} = F_d + K_f (F_d - F) \text{ 通过 } \tau = J^T F_c \text{ 得到控制力矩}$$

缺点: 对环境模型依赖强

2. 阻抗控制 Impedance Control (最常用)

不直接控制, 设计动态关系

$$M_d \ddot{x} + D_d \dot{x} + K_d (x - x_d) = F_{ext}$$

即将机器人表现为一个“质量-弹簧-阻尼”系统.

如果外界施加外力, 机器人允许产生位移.

3. Admittance Control.

阻抗控制是输出力矩, 而这里是测量力, 然后生成对应的位置命令.

f_d 为期望接触力

有力传感器, $f_s = k_s x$, $m \ddot{x} + f_s = f$ 基于弹簧形变的力测量

static equilibrium: $f_s = f_d$ (类似于纯力控制)

dynamics: $m \ddot{x} + k_s x = f_d + f_{dynamics}$